ITA0608 - Machine Learning
List of Lab Exercises

4.Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Aim

To build an Artificial Neural Network (ANN) using the Backpropagation algorithm and test it using an appropriate dataset.

Dataset

Use the **Breast Cancer Wisconsin Dataset**, which is available in **scikit-learn**, so no download is required.

Algorithm

1. Import required libraries.
2. Load the dataset.
3. Split the data into training and testing sets.
4. Normalize the data.
5. Create an Artificial Neural Network.
6. Train the model using Backpropagation.
7. Test the model.
8. Display the accuracy.

Program Code

```
from sklearn.datasets import load_breast_cancer

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.neural_network import MLPClassifier

from sklearn.metrics import accuracy_score
```

```
data = load_breast_cancer()

X = data.data

y = data.target

X_train, X_test, y_train, y_test = train_test_split(

    X, y, test_size=0.2, random_state=42

)

scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)

X_test = scaler.transform(X_test)

ann = MLPClassifier(

    hidden_layer_sizes=(10,),

    max_iter=1000,

    random_state=42

)

ann.fit(X_train, y_train)

y_pred = ann.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy :", round(accuracy * 100, 2), "%")
```

Output

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score

data = load_breast_cancer()
X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

ann = MLPClassifier(
    hidden_layer_sizes=(10,),
    max_iter=1000,
    random_state=42
)

ann.fit(X_train, y_train)

y_pred = ann.predict(X_test)

accuracy = accuracy_score(y_test, y_pred)

print("Accuracy :", round(accuracy * 100, 2), "%")
```

```
Accuracy : 98.25 %
```

Result

The Artificial Neural Network was successfully implemented using the **Backpropagation algorithm**. The model was trained and tested on the Breast Cancer dataset and achieved high classification accuracy.